

MAJOR PROJECT REPORT
ON
AI POWERED CAREER GUIDANCE-A
SCALABLE MODEL FOR PERSONALISED
RECOMMENDATIONS

Submitted By

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under the esteemed guidance of

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AUGUST 2026
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In partial fulfillment of the requirements for the award of the degree
of
Bachelor of Technology
in
Computer Science and Engineering
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APJ Abdul Kalam Technological University



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C E R T I F I C A T E

This is to certify that, the project report titled **AI POWERED CAREER GUIDANCE-A SCALABLE MODEL FOR PERSONALISED RECOMMENDATIONS**, is a bonafide record of the **CSD416 Project Phase I** presented by **ALEETTA SEBASTIAN** (Reg.No CEC23CS024), **SREELEKSHMY K S** (Reg.No CEC23CS111), **SENDHIL SELVARAJ** (Reg.No CEC23CS109), **KAVERI PS** (Reg.No CEC23CS076), Seventh Semester B. Tech. Computer Science & Engineering students, under our guidance and supervision, in partial fulfillment of the requirements for the award of the degree, B. Tech. Computer Science & Engineering of APJ Abdul Kalam Technological University.

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ABSTRACT

Career selection has become increasingly challenging due to the emergence of new technologies, interdisciplinary career opportunities, changing industry requirements, and the large amount of information available about education and career options. Existing career guidance platforms mainly function as information repositories, requiring users to manually collect, compare, and evaluate information about courses, colleges, entrance examinations, scholarships, and career opportunities. This can create confusion, lead to uninformed decisions, and result in career choices that may not align well with an individual's personality, interests, abilities, and long-term aspirations.

To address these challenges, this project proposes an AI-Assisted Personalized Career Guidance and Intelligent Decision Support Platform, designed as an explainable and data-driven recommendation system. The platform considers OCEAN (Big Five) personality traits and RIASEC (Holland) interests, along with academic performance, aptitude, interests, financial constraints, geographic preferences, aspirations, industry trends, and employability insights. It incorporates Artificial Intelligence, Machine Learning-based Recommendation, Explainable AI, Predictive Analytics, Multi-Criteria Decision Analysis (MCDA), and Knowledge-Based Recommendation to rank career pathways and provide transparent explanations for the recommendations.

The platform also integrates information related to higher education programs, universities, entrance examinations, scholarships, skills, certifications, labour-market trends, and employment opportunities. By bringing these aspects together, the system aims to reduce uncertainty, improve career awareness, and support informed career decision-making. The platform can also assist parents, mentors, and educators in providing effective career guidance.

The system is developed using Python, Django, PostgreSQL, HTML, CSS, JavaScript, and REST APIs, along with AI-based recommendation and data analytics techniques. Future extensions may include the integration of Large Language Models (LLMs), conversational AI, psychometric assessments, real-time education databases, labour-market analytics, and adaptive learning. The overall goal of the proposed platform is to support individuals in making informed career decisions by providing personalized, explainable, and data-driven career guidance.

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Chapter 1

INTRODUCTION

Choosing a suitable career has become increasingly challenging due to the rapid emergence of new technologies, evolving career opportunities, and continuously changing industry requirements. At the same time, the availability of a large volume of information related to education and careers can make it difficult for students to identify, compare, and evaluate suitable career options. An appropriate career decision therefore requires consideration of multiple factors related to an individual's interests, abilities, academic performance, aptitude, personality, and aspirations.

Existing career guidance systems often provide general or static recommendations and information related to courses, colleges, entrance examinations, scholarships, and career opportunities. However, these systems may not adequately adapt their recommendations to the individual characteristics of each student. Traditional career counselling may also consider only a limited number of factors, which can reduce the accuracy and relevance of career suggestions. Furthermore, some AI-based career recommendation systems provide limited transparency regarding the factors influencing their recommendations, which can affect user trust.

To address these challenges, the proposed AI Powered Career Guidance – A Scalable Model for Personalised Recommendations provides an AI-assisted and data-driven approach to personalized career guidance. The system considers multiple aspects of a student's profile, including personality, interests, academic performance, aptitude, and aspirations, together with relevant career and employability information. By combining these factors, the system aims to identify and rank career pathways that are better aligned with the individual characteristics of the student.

The proposed platform incorporates the OCEAN (Big Five) personality framework and the RIASEC (Holland) model as important components of the student profile. These psychological frameworks are combined with academic, aptitude, and interest-related information to develop a more comprehensive representation of the learner. The resulting profile is used to support the identification of suitable career pathways and improve the personalization of recommendations.

To improve the quality and transparency of the recommendation process, the system employs Artificial Intelligence, Machine Learning, Explainable AI (XAI), and Multi-Criteria Deci-

sion Analysis (MCDA). Machine learning and AI techniques support the recommendation process, while MCDA helps in evaluating and ranking career alternatives based on multiple relevant criteria. Explainable AI is incorporated to provide understandable explanations for the recommendations generated by the system, thereby improving transparency and helping users understand why particular career pathways are suggested.

In addition to personalized career recommendations, the platform provides access to a unified collection of career and educational information, including courses, colleges, scholarships, entrance examinations, industry trends, and employment opportunities. Integrating these resources into a single platform reduces the need for students to manually gather information from multiple sources and enables them to explore educational and career pathways in a more organized manner.

The system is intended to support students in making more informed career decisions while also assisting parents and mentors in the guidance process. By combining personal characteristics with academic information and career-related data, the platform aims to improve career awareness, reduce uncertainty, and help students make decisions that are better aligned with their interests and future opportunities.

The proposed system is developed as a scalable, data-driven career guidance and intelligent decision support platform. It brings together personalized recommendations, explainable AI, and comprehensive career-related information within a unified system. The overall objective is to provide students with a more informed, personalized, and transparent approach to career exploration and decision-making.

Future extensions of the system may include the integration of Large Language Models (LLMs), conversational AI, more comprehensive psychometric assessments, real-time education databases, labour-market analytics, and adaptive learning capabilities. These enhancements could further improve the platform's ability to provide personalized, interactive, and up-to-date career guidance.

Chapter 2

PROBLEM STATEMENT

2.1 Problem Statement

Career guidance plays an important role in helping students identify suitable career pathways based on their interests, abilities, and future goals. However, many existing career guidance systems provide static and generic recommendations that do not sufficiently adapt to the individual needs of students. Traditional career counselling approaches also tend to consider only a limited number of factors when suggesting career options, which can result in less accurate or unsuitable recommendations.

Another major limitation of existing AI-based career recommendation systems is the lack of transparency and explainability. When recommendations are generated without clearly explaining the factors behind them, users may find it difficult to understand or trust the suggested career options. This reduces the effectiveness of AI-based systems as decision-support tools.

Furthermore, existing systems do not effectively combine psychometric assessments, such as the RIASEC model and personality assessment, with AI-based techniques for personalized career prediction. As a result, important information about a student's interests and personality may not be adequately incorporated into the recommendation process.

Therefore, there is a need for a scalable, intelligent, and data-driven career guidance platform that can provide personalized and accurate career recommendations. Such a system should consider individual interests and relevant career-related factors while also providing understandable recommendations that can support better career decision-making and alignment with employability.

2.2 Objective

The main objective of the proposed system is to develop an AI-based career guidance system for students that can provide personalized recommendations and support informed career decision-

making. The system aims to overcome the limitations of generic career guidance by considering individual factors and providing understandable recommendations.

1. To develop an AI-based career guidance system for students that assists in identifying suitable career options using AI-based recommendation techniques.
2. To provide personalized career recommendations based on individual profiles, considering relevant individual factors to generate career suggestions that are better aligned with the student's needs and interests.
3. To help students choose suitable courses and career paths with confidence by providing relevant career guidance and information to support their decision-making process.
4. To use Explainable Artificial Intelligence (XAI) to make recommendations easy to understand, by providing clear explanations for the career recommendations generated by the system.
5. To improve career decision-making by reducing confusion and increasing awareness of available career options, courses, and pathways, thereby helping students make more informed decisions.

Chapter 3

LITERATURE SURVEY

3.1 [1] AI-Powered Career Guidance: A Scalable Model for Personalized Recommendations

Authors: Minakshi Roy and Kalpana Sharma

Published On: 2026

- **About:** Proposes an AI-based career guidance framework combining the RIASEC model with machine learning for personalized career recommendations.
- **Methodology:** Uses RIASEC profiles and a career dataset with Decision Tree, SVM, and XGBoost to predict suitable career pathways.
- **Advantages:** Provides personalized, data-driven recommendations with improved prediction accuracy.
- **Disadvantages:** Uses older 2016 data and mainly relies on self-reported interests, with limited consideration of other factors and real-time adaptation.

3.2 [2] PCRS: Personalized Career-Path Recommender System for Engineering Students

Authors: Manar Qamhieh, Haya Sammaneh, and Mona Nabil Demaidi

Published On: 2020

- **About:** PCRS uses AI and fuzzy logic to recommend suitable engineering disciplines based on academic performance, personality, gender, and extracurricular skills.
- **Methodology:** Uses STEM grades, MBTI personality type, and extracurricular interests with a Mamdani fuzzy-logic system to rank engineering disciplines.
- **Advantages:** Provides personalized recommendations by considering multiple academic, personality, and extracurricular factors.
- **Disadvantages:** Has limited agreement with actual disciplines, a small evaluation sample, and mainly focuses on engineering fields.

3.3 [3] Artificial Intelligence-Powered Personalised Career Guidance System

Authors: Victor Emmanuel Kulugh, Kwembe Prince Aondover, and Ageebie Silas Faki

Published On: 2025

- **About:** AI-powered system that provides personalized career recommendations based on users' profiles and aspirations.
- **Methodology:** Uses the Waterfall methodology, system modelling diagrams, Supabase for database and authentication, and an AI-based recommendation layer.
- **Advantages:** Provides tailored career suggestions, accurate recommendations, efficient execution, and positive user feedback.
- **Disadvantages:** Requires continuous updates and refinement, and further development is needed to improve the system.

3.4 [4] Mining Campus Big Data: Prediction of Career Choice Using Interpretable Machine Learning Method

Authors: Yuan Wang, Liping Yang, Jun Wu, Zisheng Song, and Li Shi

Published On: 2022

- **About:** Presents an interpretable machine-learning framework using XGBoost and SHAP to predict and explain students' career choices.
- **Methodology:** Uses educational records of 18,000 students, with XGBoost for prediction, TPE for optimization, and SHAP for interpreting important career-choice factors.
- **Advantages:** Provides accurate and interpretable career-choice predictions, with SHAP helping identify influential factors.
- **Disadvantages:** Uses data from a specific college and mainly supports educational institutions rather than providing complete personalized guidance to students.

3.5 [5] Career Choice Prediction Based on Campus Big Data—Mining the Potential Behavior of College Students

Authors: Min Nie, Zhaohui Xiong, Ruiyang Zhong, Wei Deng, and Guowu Yang

Published On: 2020

- **About:** Presents the ACCBOX model using campus behavioral data to predict students' career choices.
- **Methodology:** Uses behavioral data from 4,634 students, with clustering, feature engineering, behavioral entropy, and XGBoost for career prediction.
- **Advantages:** Improves career-choice prediction by combining behavioral information, cluster centers, and XGBoost.
- **Disadvantages:** Mainly relies on campus behavioral data and does not include academic or questionnaire information.

3.6 [6] Investigating Machine Learning's Capacity to Enhance the Prediction of Career Choices

Authors: Q. Chelsea Song, Hyun Joo Shin, Chen Tang, Alexis Hanna, and Tara Behrend

Published On: 2024

- **About:** Examines the use of machine learning to improve career-choice prediction based on vocational interests.
- **Methodology:** Uses data from 81,267 participants and compares traditional interest-profile matching with machine-learning models using RIASEC interest scores.
- **Advantages:** The ML-based approach provides higher prediction accuracy and can use both broad and specific career interests.
- **Disadvantages:** May underpredict low-frequency career categories, and broad RIASEC categories may not capture specific career preferences.

3.7 [7]Artificial Intelligence in Personalized Learning for Career Guidance: A Systematic Literature Review

Authors: Wa Ode Husniah, Unhaluddin T. Kurniawan, and Maria Ulfa

Published On: 2025

- **About:** Reviews the use of AI technologies such as machine learning, chatbots, and recommender systems for personalized career guidance.
- **Methodology:** Follows the PRISMA 2020 framework and analyses 18 studies selected from four major databases.
- **Advantages:** Provides an overview of AI-based career guidance and highlights improvements in career clarity, self-efficacy, and decision-making.
- **Disadvantages:** Faces limitations related to database and language scope, along with privacy, bias, ethical issues, and educator readiness.

3.8 [8]Artificial Intelligence in Career Counseling: A Test Case with ResumAI

Authors: Muhammad Rahman, Sachi Figliolini, Joyce Kim, Eivy Cedeno, Charles Kleier, Chirag Shah, and Aman Chadha

Published On: 2023

- **About:** Examines AI-based career counselling through ResumAI, a web-based chatbot that provides personalized career and resume feedback.
- **Methodology:** Uses the OpenAI API to generate responses based on users' resumes and career-related questions through a web interface.
- **Advantages:** Provides quick, personalized, and accessible career and resume feedback.
- **Disadvantages:** Has limitations in fine-tuning and up-to-date information, along with concerns related to bias, explainability, ethics, and user authentication. .

3.9 [9]AI-Powered Career Guidance: Personalizing Career Paths with Machine Learning

Authors: Shivangini Bihari and Md. Irfan Alam

Published On: 2025

- **About:** Discusses the use of AI and machine learning for personalized career guidance based on skills, interests, personality, values, job-market trends, and salary information.
- **Methodology:** Reviews machine learning, recommendation systems, deep learning, NLP, and chatbot approaches used in career guidance.
- **Advantages:** AI and ML can provide personalized recommendations and make career exploration more efficient
- **Disadvantages:** Model effectiveness depends on data quality and availability, while real-time data may be needed to keep career information updated..

3.10 [10]A Review on AI-Based Chatbots for Personalized Career Guidance

Authors: Monika D. S. and Dr. K. Balaji

Published On: 2025

- **About:** Reviews an AI-based career counselling system that uses students' education, skills, interests, and preferences for personalized career recommendations
- **Methodology:** Uses data collection, machine learning, and chatbot deployment with algorithms such as KNN, Random Forest, SVM, and Decision Tree, along with spaCy and BERT.
- **Advantages:** Provides personalized recommendations, accessible chatbot interaction, and real-time career guidance.
- **Disadvantages:** Multiple AI components increase implementation complexity and require careful handling of personal and emotional data..